Supplementary file 7-Thematic synthesis of empirical research and case studies as described by Thomas and Harden

Final codes	Descriptive themes	Analytical themes
1. Unpredictability leading to uncertainties	1. Stochastic uncertainty	**1. Aleatoric uncertainty**
2. Statistical uncertainty		
3. Future risk leading to uncertainties		
4. Expensive investigations	2. Diagnostic uncertainty	**2. Clinical uncertainty**
5. Lack of robust clinical protocols		
6. Resemblance to other diseases		
7. Role of co-factors	3. Prognostic uncertainty	
8. Seriousness of condition		
9. Negotiation skills	4. Communication of knowledge	**3. Knowledge exchange related uncertainty**
10. Uncertainties due to cues picked during conversation		
11. Uncertainties due to language used during communication		
12. Uncertainty due to lack of patient-centred communication		
13. Clinicians' ignorance or lack of skill	5. Knowledge related to stakeholders	**4. Epistemic uncertainty**
14. Patient's level of education		
15. Ambiguity leading to uncertainties	6. Quality of knowledge or information	
16. Complexity leading to uncertainties		
17. Lack of credibility leading to uncertainties		
18. Growth of medical technology related knowledge leading to uncertainties	7. Quantity of knowledge	
19. Less knowledge or information leading to uncertainties		
20. More knowledge or information leading to uncertainties		
21. Prioritising patients due to lack of resources	8. Ethical dilemmas	**5. Ethical uncertainty**
22. Time pressure dictating patient treatment		
23. To inform or not to inform patients about lack of evidence		
24. Clinicians making decisions on behalf of the patient or family members	9. Proxy uncertainties	
25. Parents making decisions on behalf of their child		
26. Suddenness of the situation		**6. Global uncertainty**

Final codes	Descriptive themes	Analytical themes
27. Uncertainty due to lack of conclusive results from emergency research	10. Lack of evidence in new diseases (Pandemics/epidemics))	
28. Uncertainties created by media	11. Uncertainties due to (global) stakeholders involved in key decision making	
29. Uncertainties due to social arrangement of power		
30. Uncertainties due to subjective decision making		
31. Inability to establish a cause-and-effect relationship	12. Uncertainties which are disease related (global emergency related)	
32. Uncertainties due to lack of pattern in disease presentation		
33. Timing of the week or day	13. Contextual uncertainties	**7. Health care system uncertainty**
34. Errors in reporting	14. Investigative uncertainties	
35. Over testing in response to or leading to uncertainties		
36. Lack of check mechanism	15. Lack of resources	
37. Lack of facilities		
38. Level of experience	16. Uncertainties related to doctors	**8. Personal uncertainty**
39. Clinicians' personal fears		
40. Previous experiences (clinician)		
41. Proxy uncertainties (clinician)		
42. Risk taking behaviour (Clinician)		
43. Uncertainty intolerance (Clinician)		
44. Unclear patient characteristics	17. Uncertainties related to patients	
45. Influence of friends and family		
46. Patient's level of education		
47. Patients' personal beliefs		
48. Patients' personal fears		
49. Risk taking behaviour		
50. Trust issues		
51. Uncertainty negatively perceived		
52. Uncertainty positively perceived (coping with)		
53. Proxy uncertainties (Carers)	18. Uncertainties related to carers	
54. Associated fears of parents		
55. Personal interpretations (parents)		
56. Lack of quantifiability or measurability	19. Methodological uncertainties	**9. Parameter uncertainty**

Final codes	Descriptive themes	Analytical themes
57. Model inadequacy		
58. Procedural uncertainties		
59. Guidelines cannot be generalised	20. Uncertainty pertaining to guidelines	
60. Inability to apply guidelines to all situations		
61. Uncertainties due to socio-economic inequalities and inequities	21. Uncertainties in establishing and implementing the national standards	**10. Public health uncertainty**
62. Uncertainties in implementing preventive measures		
63. Uncertainty due to lack of standardised definitions		
64. Uncertainties due to challenges in data collection	22. Uncertainties related to epidemiological data	
65. Uncertainties due to lack of previous records		
66. Uncertainty due to unclear risk distribution		
67. Interaction of clinicians with other health care team leading to uncertainties	23. Interpersonal relationship leading to uncertainties	**11. Relational uncertainty**
68. Interaction or conflict between patients' friends, relatives with clinician team leading to uncertainties		
69. Physician- patient interaction leading to uncertainties		
70. Improper team management leading to uncertainties	24. Lack of unified teamwork leading to uncertainties	
71. Trust issues in the team leading to uncertainties		
72. Uncertainties due to avoiding referral	25. Networking leading to uncertainties	
73. Uncertainties due to involvement of multiple people		
74. Uncertainties due to referral		
75. Uncertainties due to seeking help from others		
76. Fear of stigma leading to uncertainties	26. Social relationships leading to uncertainties	
77. Helplessness or feeling unsupported leading to uncertainties		